```
import numpy as np
import pandas as pd

import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import r2_score
```

```
import numpy as np
import pandas as pd

X = np.array([1.1, 1.3, 1.5, 2.0, 2.2, 2.9, 3.0, 3.2, 3.7, 4.0, 2.9])

Y = np.array([39000, 46000, 37700, np.nan, 39800,
              56600, 60100, 54400, 57100, 63200, 56600])

df = pd.DataFrame({
    "Experience": X,
    "Salary": Y
})

print(df)
```

```
    Experience   Salary
0          1.1  39000.0
1          1.3  46000.0
2          1.5  37700.0
3          2.0      NaN
4          2.2  39800.0
5          2.9  56600.0
6          3.0  60100.0
7          3.2  54400.0
8          3.7  57100.0
9          4.0  63200.0
10         2.9  56600.0
```

```
plt.plot(X,Y,marker="D")
plt.xlabel("Years of Experience")
plt.ylabel("Salary")
plt.title("Original Data - Experience vs Salary")
plt.grid()
plt.show()
```




```
df.isnull()
```

	Experience	Salary
0	False	False
1	False	False
2	False	False
3	False	True
4	False	False
5	False	False
6	False	False
7	False	False
8	False	False
9	False	False
10	False	False

```
print("\nMissing Values:")
print(df.isnull().sum())
```

```
Missing Values:
Experience    0
Salary        1
dtype: int64
```

```
print("\nDuplicate Rows:")
print(df.duplicated().sum())
```

```
Duplicate Rows:
1
```

```
df["Salary"] = df["Salary"].fillna(df["Salary"].mean())
df
```

	Experience	Salary
0	1.1	39000.0
1	1.3	46000.0
2	1.5	37700.0
3	2.0	51050.0
4	2.2	39800.0
5	2.9	56600.0
6	3.0	60100.0
7	3.2	54400.0
8	3.7	57100.0
9	4.0	63200.0
10	2.9	56600.0

Next steps: Generate code with df | New interactive sheet

```
df = df.drop_duplicates()
df
```

	Experience	Salary
0	1.1	39000.0
1	1.3	46000.0
2	1.5	37700.0
3	2.0	51050.0
4	2.2	39800.0
5	2.9	56600.0
6	3.0	60100.0
7	3.2	54400.0
8	3.7	57100.0
9	4.0	63200.0

Next steps: Generate code with `df` New interactive sheet

```
print(df.isnull().sum())
print(df.duplicated().sum())
```

```
Experience    0
Salary        0
dtype: int64
0
```

```
plt.plot(X, Y, marker='D')

plt.xlabel("Years of Experience")
plt.ylabel("Salary")
plt.title("Original Data - Experience vs Salary")
plt.grid()

plt.show()
```




```
X = df[["Experience"]]
Y = df["Salary"]

X_train, X_test, Y_train, Y_test = train_test_split(
    X, Y, test_size=0.2, random_state=1
)
```

```
X_test.shape
```

```
(2, 1)
```

```
Y_test.shape
```

```
(2,)
```

```
X_train.shape
```

```
(8, 1)
```

```
Y_train.shape
```

```
(8,)
```

```
model=LinearRegression()
```

```
model.fit(X_train,Y_train)
```

▾ LinearRegression

```
LinearRegression()
```

```
Y_pred = model.predict(X_test)
print("prediction value", Y_pred)
```

```
prediction value [44018.41342171 61553.10687655]
```

```
r2 = r2_score(Y_test, Y_pred)

print("R2 Score:", r2)
```

```
R2 Score: 0.8688670353628323
```

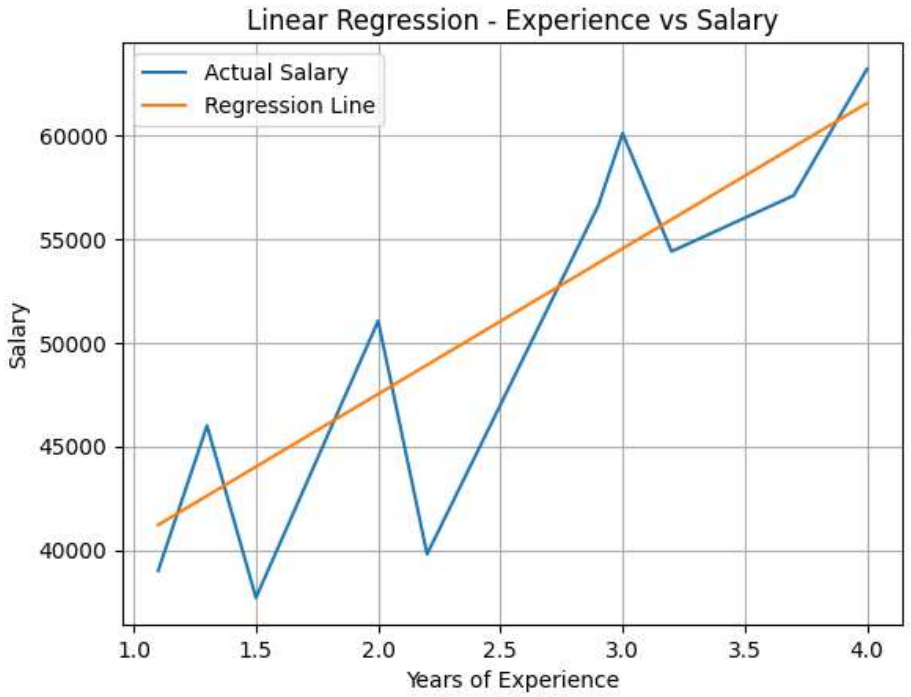
```
experience=np.array([[3]])
salary=model.predict(experience)
print("Predicted Salary for 5 years of Experience:", salary[0])
```

```
Predicted Salary for 5 years of Experience: 54539.229494614745
/usr/local/lib/python3.13/dist-packages/sklearn/utils/validation.py:2739: UserWarning: X does not have valid feature names, but LinearRegression w
  warnings.warn(
```

```
plt.plot(X,Y,label="Actual Salary")
plt.plot(X, model.predict(X),label="Regression Line")

plt.xlabel("Years of Experience")
plt.ylabel("Salary")
plt.title("Linear Regression - Experience vs Salary")

plt.legend()
plt.grid()
plt.show()
```




```
df.mean()
```

	0
Experience	2.49
Salary	50495.00

dtype: float64

```
df.median()
```

	0
Experience	2.55
Salary	52725.00

dtype: float64

```
df.std()
```

	0
Experience	1.018114
Salary	9312.848055

dtype: float64

```
df.corr()
```